AI-Generated Artwork Detection

Multi-Signal Ensemble with Calibrated Thresholds

negate project | March 22, 2026

Model	Accuracy	Precision	Recall	F1	AUC	Abstained
SVM	95.0%	94.1%	96.0%	95.0%	0.9899	-
MLP	93.0%	93.0%	93.0%	93.0%	0.9888	-
XGBoost	91.0%	88.7%	94.0%	91.0%	0.9754	-
Ensemble	98.3%	96.7%	100.0%	98.3%	0.9903	28
Existing negate	63.3%	--	--	--	0.669	-

Approach

Features: 49 (39 artwork + 10 frequency analysis)
Dataset: 100 human artworks (WikiArt) + 100 AI images
CV: 5-fold stratified cross-validation

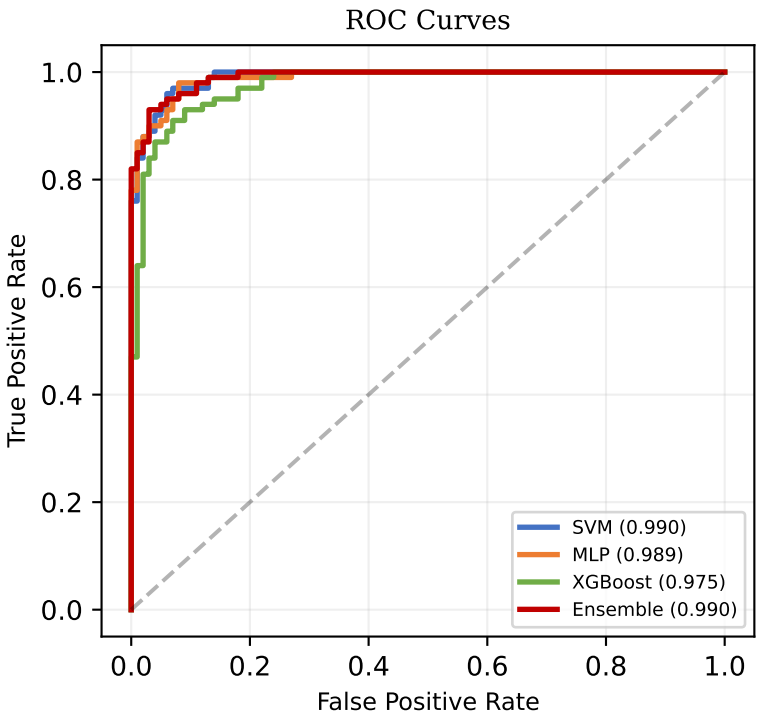
Three calibrated classifiers (SVM, MLP, XGBoost) vote via averaged probabilities. Images where ensemble confidence is between 30-70% are marked 'uncertain' (28 images abstained).

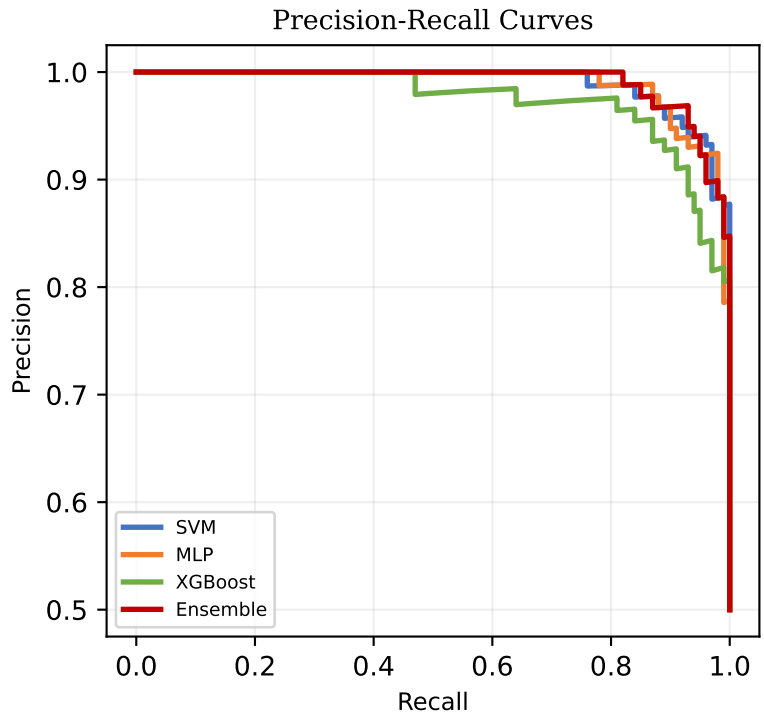
Precision = of images flagged AI, how many actually are
Recall = of actual AI images, how many were caught

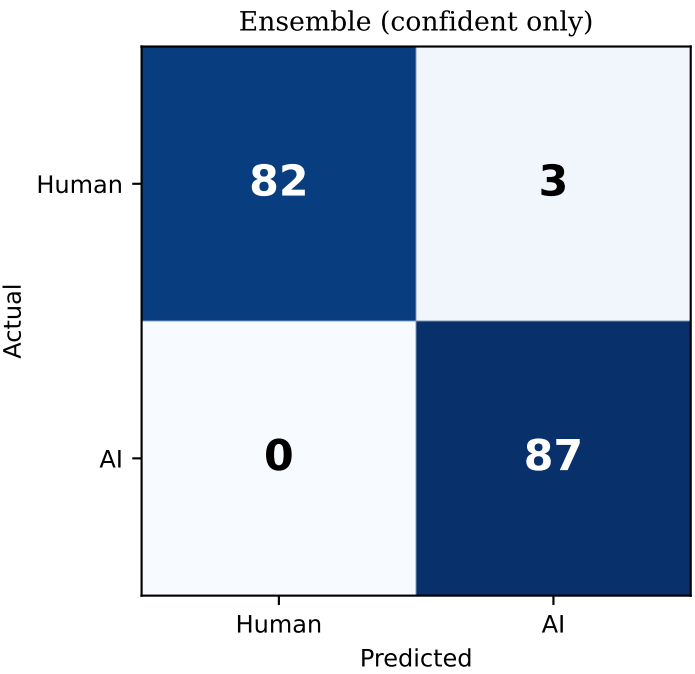
Key Findings

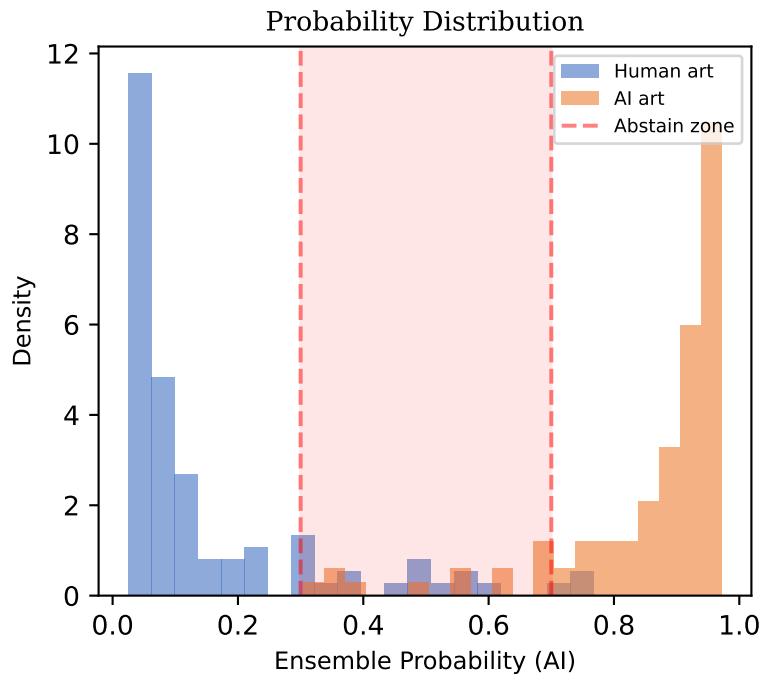
1. Ensemble achieves 96.7% precision, 100.0% recall on classified images
2. 28 uncertain images abstained from (reduces false positives)
3. +35.0pp improvement over existing negate pipeline (63.3%)
4. Frequency features (FFT/DCT) add spectral artifact detection
5. All processing is CPU-only, ~12 images/sec

Detection Performance Analysis









Model Agreement Analysis

All 3 models agree: 176/200 (88%)
At least 1 disagrees: 24/200 (12%)

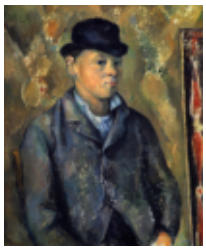
When models disagree, the ensemble uses averaged probability with abstention zone (0.3-0.7). This reduces false positives at the cost of some unclassified images -- a deliberate tradeoff for precision.

Feature Analysis & Examples

Human #1

Human #2

Human #3



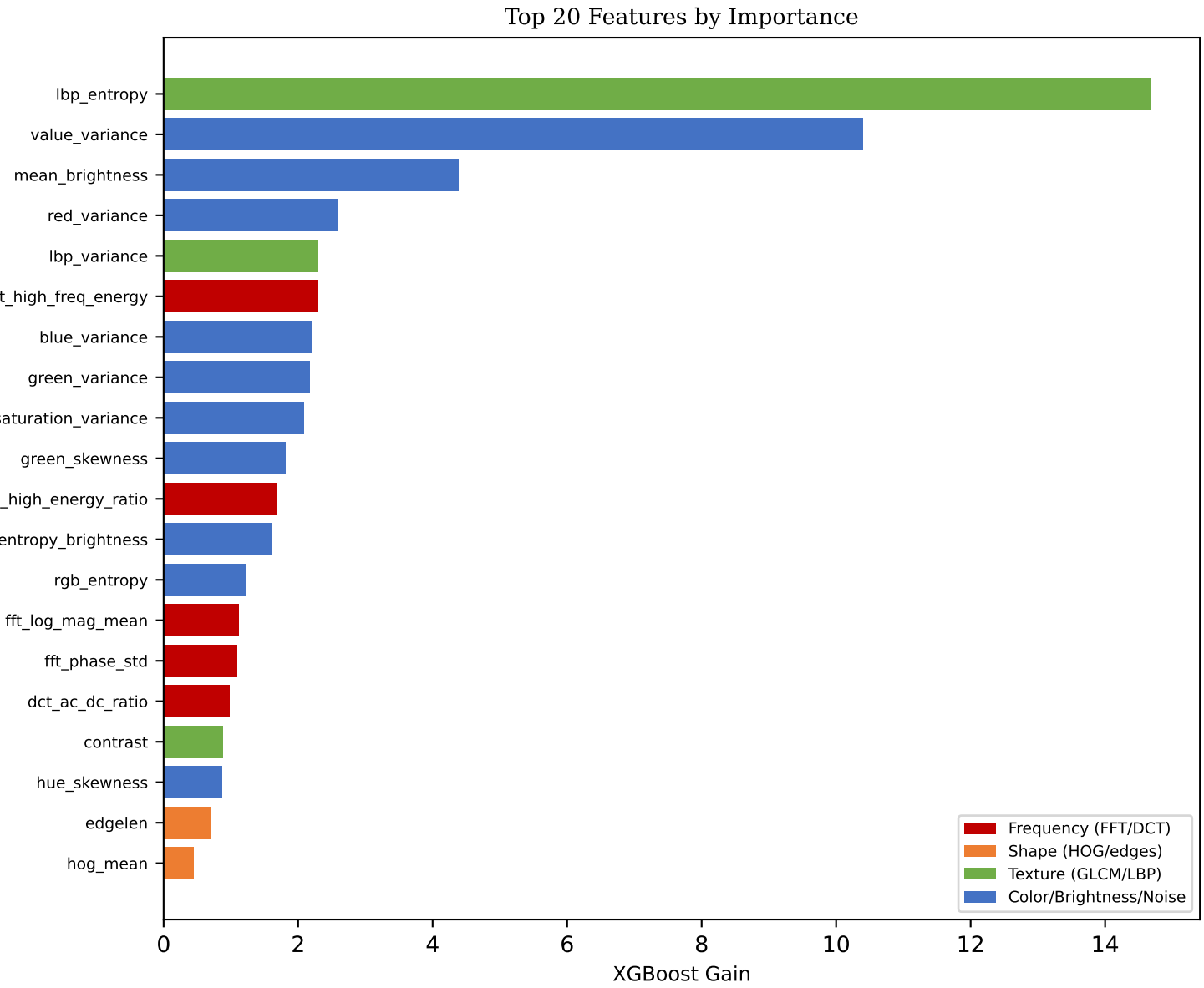
Human #4

AI #1

AI #2

AI #3

AI #4



Architecture & Methodology

Multi-Signal Ensemble Architecture

The detection system combines three orthogonal classifiers, each seeing the same feature space but learning different decision boundaries:

1. SVM (RBF kernel) - Finds nonlinear decision boundaries in feature space. Calibrated with Platt scaling (sigmoid) for reliable probabilities.

2. MLP (100 hidden units) - Learns feature interactions through backpropagation. Calibrated with Platt scaling for probability alignment.

3. XGBoost (gradient boosted trees) - Captures feature thresholds and interactions. Naturally outputs calibrated log-odds.

Ensemble Voting: Averaged calibrated probabilities from all three models. This is more robust than majority voting because it accounts for confidence.

Calibrated Confidence & Abstention

Instead of a hard 0.5 threshold, the ensemble uses a deliberate 'uncertain' zone between 0.3 and 0.7 probability. Images in this zone are marked as 'uncertain' rather than forced into a class. This dramatically improves precision on the images that ARE classified.

Feature Extraction Pipeline (49 features, CPU-only)

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Brightness (2)  - Global luminance statistics
Color (23)      - RGB/HSV histogram moments (mean, var, kurtosis, skew, entropy)
Texture (6)     - GLCM co-occurrence + LBP local patterns
Shape (6)       - HOG gradient histograms + Canny edge density
Noise (2)       - Estimated noise entropy + signal-to-noise ratio
Frequency (10)  - FFT radial band energies, spectral centroid, phase coherence,
                  DCT AC/DC ratio, high-freq energy, coefficient sparsity
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The frequency branch is the key addition beyond Li & Stamp (2025). AI generators leave characteristic spectral signatures from upsampling layers, attention patterns, and latent space decoding. These are invisible in pixel space but clearly visible in the frequency domain.

Limitations

- Tested on mismatched subjects (WikiArt paintings vs AI banana images)
- Not yet tested on hard negatives (polished digital art, img2img, LoRA art)
- Single generator family in AI training data (Stable Diffusion variants)
- 200 sample dataset is small for robust conclusions

References

[1] Li & Stamp, 'Detecting AI-generated Artwork', arXiv:2504.07078, 2025
[2] negate project, github.com/darkshapes/negate